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Qv Help Search

```
> # =====
> # IMPORT DATASET
> # =====
>
> data <- read.csv(file.choose(), stringsAsFactors = FALSE)
>
> # Display dataset
> print(data)
Student_ID Gender Age Course Study_Time..hrs. Videos_Watched Quiz_Score
1      L01   Male  20      R           3.5           12          78
2      L02 Female  22      R           4.2           15          85
3      L03   Male  19    SQL           2.0            8          65
4      L04 Female  21      R           5.0           18          92
5      L05   Male  23      R           2.5            9          70
6      L06 Female  20    SQL           4.0           14          88
Login_Date
1 2025-01-05
2 2025-01-05
3 2025-02-08
4 2025-02-08
5 2025-03-12
6 2025-03-12
>
> # =====
> # 1. HISTOGRAM OF QUIZ SCORE
> # =====
>
>
> quartz()
>
> hist(data$Quiz_Score,
+       main = "Distribution of Quiz Scores",
+       xlab = "Quiz Score",
+       ylab = "Frequency",
+       col = "skyblue",
+       border = "black")
>
> # =====
> # 1. BOXPLOT OF QUIZ SCORE BY COURSE
> # =====
>
>
> quartz()
>
> boxplot(Quiz_Score ~ Course,
+         data = data,
+         main = "Quiz Score by Course",
+         xlab = "Course",
+         ylab = "Quiz Score",
+         col = c("lightgreen", "lightpink"))
>
> # =====
> # 2. SCATTER PLOT (BUBBLE CHART)
> # =====
>
```



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> # =====
> # 2. SCATTER PLOT (BUBBLE CHART)
> # =====
>
> quartz()
>
> plot(data$Study_Time,
+       data$Quiz_Score,
+       cex = data$Videos_Watched/5,
+       pch = 19,
+       col = rgb(0, 0, 1, 0.5),
+       main = "Study Time vs Quiz Score",
+       xlab = "Study Time (Hours)",
+       ylab = "Quiz Score")
>
> # =====
> # 3. CONVERT LOGIN_DATE TO DATE FORMAT
> # =====
>
> data$Login_Date <- as.Date(data$Login_Date,
+                             format = "%Y-%m-%d")
>
> # Extract Month
> data$Month <- format(data$Login_Date, "%Y-%m")
>
> # =====
> # CALCULATE AVERAGE QUIZ SCORE PER MONTH
> # =====
>
> avg_score <- aggregate(Quiz_Score ~ Month,
+                         data = data,
+                         FUN = mean)
>
> print(avg_score)
      Month Quiz_Score
1 2025-01         81.5
2 2025-02         78.5
3 2025-03         79.0
>
> # =====
> # LINE CHART WITH MOVING AVERAGE
> # =====
>
> quartz()
>
> plot(1:nrow(avg_score),
+      avg_score$Quiz_Score,
+      type = "o",
+      pch = 16,
+      xaxt = "n",
+      main = "Average Quiz Score per Month",
+      xlab = "Month",
+      ylab = "Average Quiz Score")
```

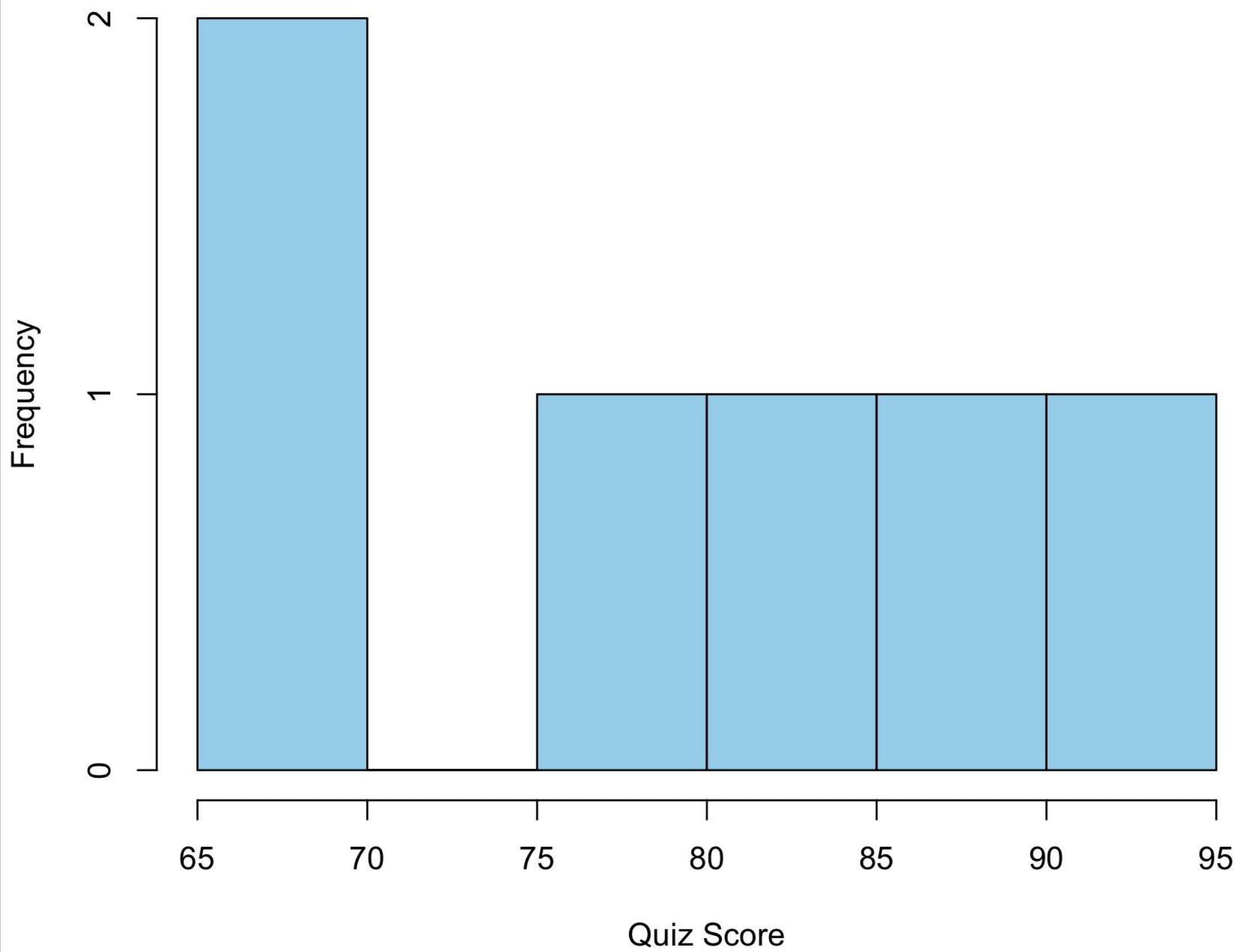


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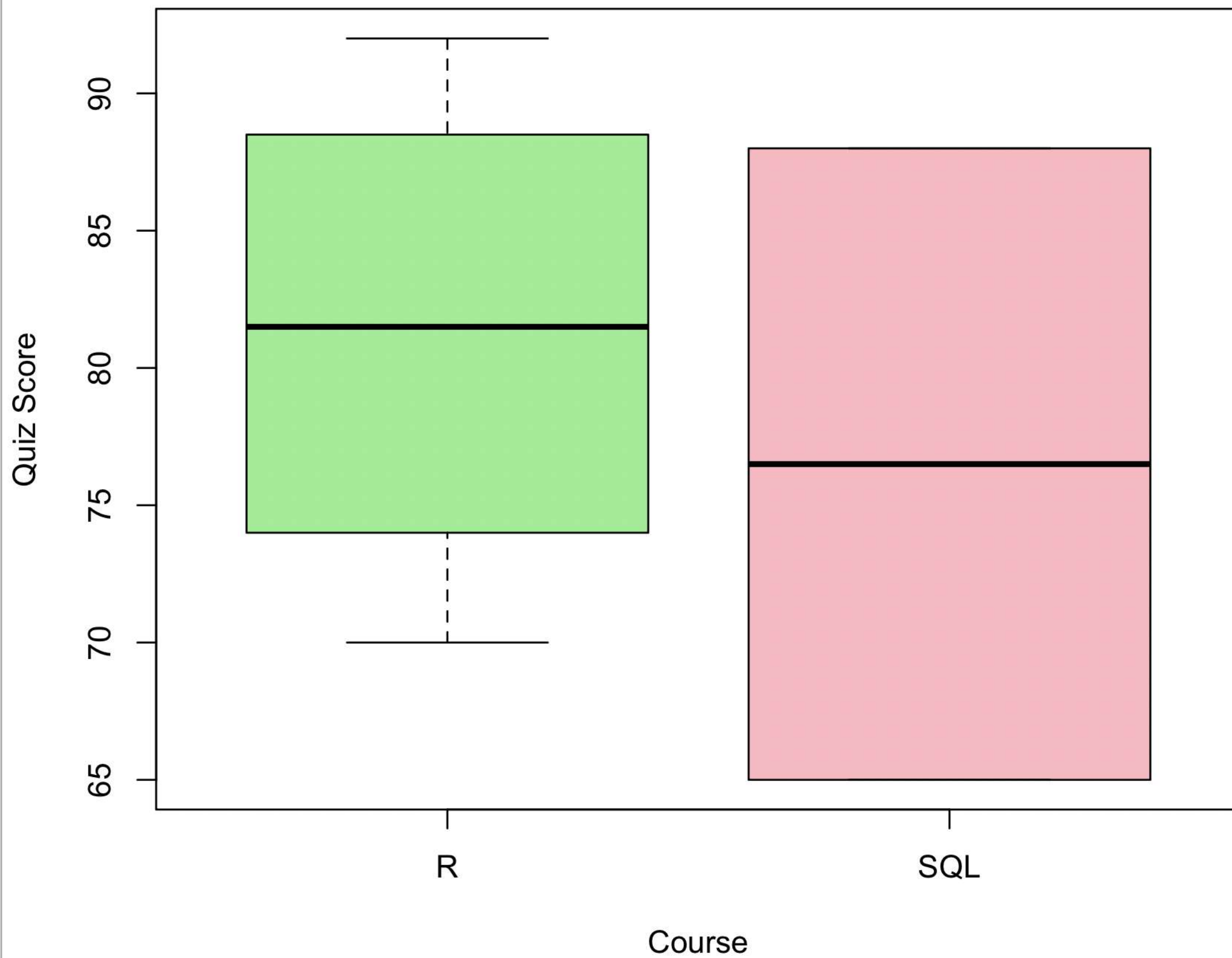
Qv Help Search

```
+                               format = "%Y-%m-%d")
>
> # Extract Month
> data$Month <- format(data$Login_Date, "%Y-%m")
>
> # =====
> # CALCULATE AVERAGE QUIZ SCORE PER MONTH
> # =====
>
> avg_score <- aggregate(Quiz_Score ~ Month,
+                         data = data,
+                         FUN = mean)
>
> print(avg_score)
  Month Quiz_Score
1 2025-01      81.5
2 2025-02      78.5
3 2025-03      79.0
>
> # =====
> # LINE CHART WITH MOVING AVERAGE
> # =====
>
> quartz()
>
> plot(1:nrow(avg_score),
+      avg_score$Quiz_Score,
+      type = "o",
+      pch = 16,
+      xaxt = "n",
+      main = "Average Quiz Score per Month",
+      xlab = "Month",
+      ylab = "Average Quiz Score")
>
> axis(1,
+      at = 1:nrow(avg_score),
+      labels = avg_score$Month)
>
> # Moving Average Smoothing
> moving_avg <- stats::filter(avg_score$Quiz_Score,
+                             rep(1/2, 2),
+                             sides = 1)
>
> lines(1:nrow(avg_score),
+      moving_avg,
+      col = "red",
+      lwd = 2)
>
> legend("topleft",
+      legend = c("Average Score", "Moving Average"),
+      col = c("black", "red"),
+      lty = c(1,1),
+      pch = c(16, NA))
>
```

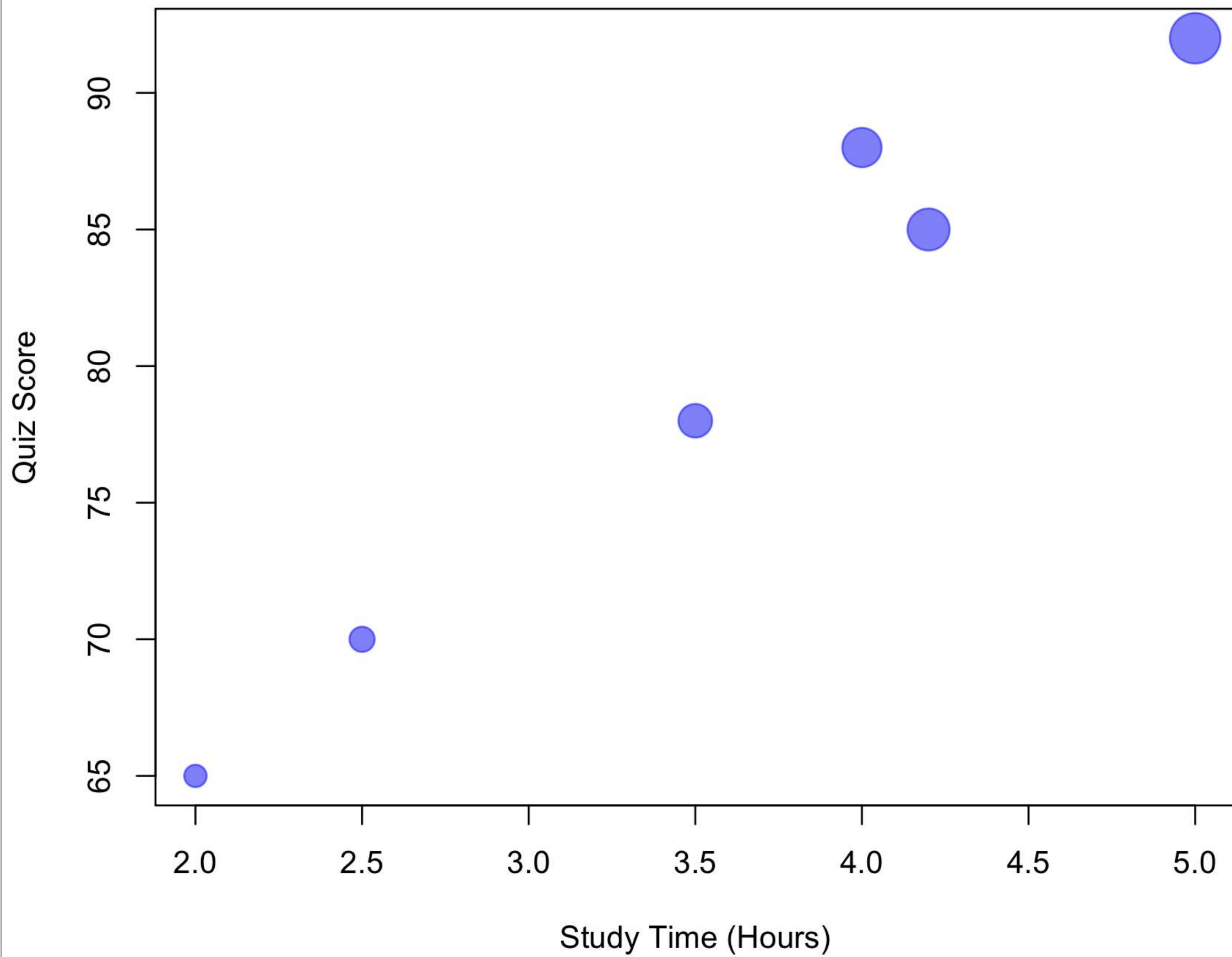

Distribution of Quiz Scores



Quiz Score by Course

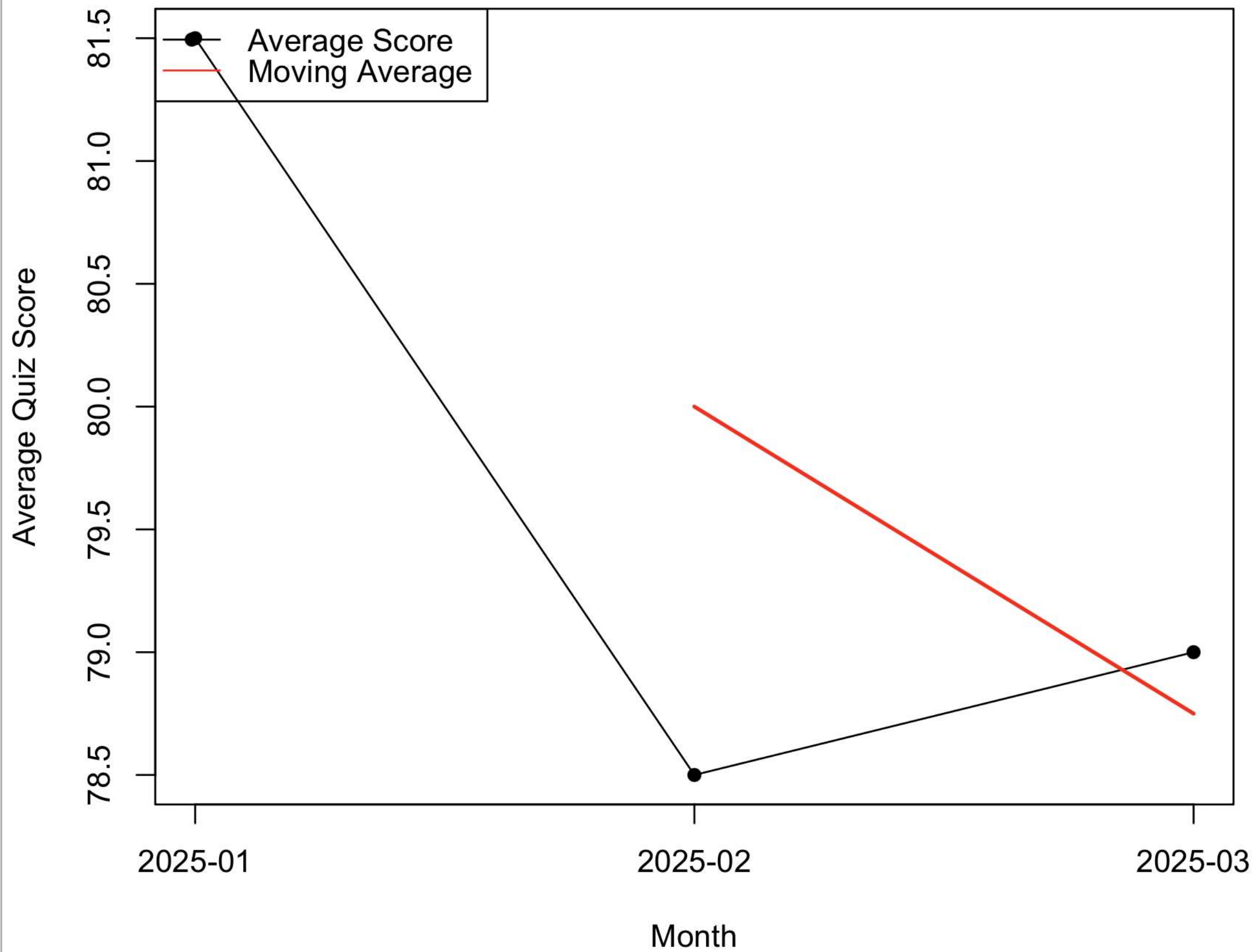


Study Time vs Quiz Score





Average Quiz Score per Month



Standard

Show Me

Data Analytics

Online_Learning_Activity

Search

Tables

Abc Course

Abc Gender

Login Date

Abc Student ID

Abc Measure Names

Age

Quiz Score

Study Time (hrs)

Videos Watched

Online_Learning_Activity....

Measure Values

Pages

Columns SUM(Study Time (hr..

Rows SUM(Quiz Score)

Filters

Action (Course)

Marks

Circle

Color

Size

Label

Detail

Tooltip

Student ID

